Appendix B
Model Specifications for an ABM of Brokerage in Matching Markets

This document specifies an agent-based model of brokered matching.

The model is a discrete-time simulation of a matching market with *principals* and a *broker*. Principals seeking pairwise matches either search on their own or outsource to the broker. All economic quantities (match output, fees, surplus) are in the same payoff units. All principals use heuristic decision rules. No principal holds beliefs about other principals' strategies.

The model specification covers principals (§1), the matching problem (§2), how principals learn to predict match value (§3), match economics (§4), network structure and principal turnover (§5), search and broker access (§6), the outsourcing decision (§7), and period measures (§8). Baseline parameter values and reproducibility are documented in Part II (§9).

Part I. Model Specification

1. Principals and Time

The model runs for T discrete periods (default 500). It has N principals (default 1000), a single broker, and an undirected network G over the principals and broker.

Each principal i is characterized by:

- **Type** $\mathbf{x}_i \in \mathbb{R}^d$: a fixed vector of observable characteristics, drawn as specified below and used by the matching function (§2).
- **Current-period matches** M_i^t: active bilateral relationships involving i in period t. Each counterparty can appear at most once in M_i^t; incoming accepted offers are not capped.
- **Active demand bound** K: the maximum number of active demands a principal can initiate in a period (default $K = 5$; §6).
- **Experience history** $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}$: (other party's type, realized match output) pairs from all matches involving i (§3b).
- **Satisfaction indices** $s_{i,c}^t$: one scalar per search channel $c \in \{\text{self}, \text{broker}\}$, used in the outsourcing decision (§7).

Principal Types

Type vectors are multidimensional but concentrated near a smooth one-dimensional curve on the unit sphere in $\mathbb{R}^d$. Let $u \in [0, 1]$ index position along the curve:

$$\boldsymbol{\gamma}(u) = \frac{\tilde{\boldsymbol{\gamma}}(u)}{\|\tilde{\boldsymbol{\gamma}}(u)\|}, \qquad \tilde{\gamma}_\ell(u) = \begin{cases} \sin(2\pi\nu_\ell u + \psi_\ell) & \ell = 1, \ldots, d_\gamma \\ 0 & \ell = d_\gamma + 1, \ldots, d. \end{cases}$$

The frequencies $\nu_\ell \sim \text{Unif}\{1, 2, 3, 4, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ are drawn once per simulation. The parameter $d_\gamma \leq d$ is the number of coordinates in which the curve varies.

Each principal is drawn at a random position $u_i \sim \text{Unif}[0, 1]$ on the curve, then perturbed:

$$\mathbf{x}_i = \frac{\boldsymbol{\gamma}(u_i) + \boldsymbol{\epsilon}_i}{\|\boldsymbol{\gamma}(u_i) + \boldsymbol{\epsilon}_i\|}, \qquad \boldsymbol{\epsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d}\mathbf{I}_d\right).$$

Noise is applied in all d dimensions, then the vector is re-normalized. The scale $\sigma_x/\sqrt{d}$ keeps the expected displacement from the curve approximately σ_x for any d.

Thus the curve remains one-dimensional for every d_γ. Increasing d_γ changes how many coordinates contain systematic curve variation; the remaining coordinates contain only perturbation noise.

Broker

The broker is a permanent node in G and has no type vector. Its state is:

- **Experience history** $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}$: (party 1 type, party 2 type, realized match output) triples from mediated matches (§3c). The party labels have no directional meaning because the broker's pair features are symmetric.
- **Roster** Roster^t: the broker's standing access base (§6b).
- **Current clients** D^t: principals who outsource to the broker in period t (§6b).
- **Reputation** rep_b^t: the average broker satisfaction of current clients (§7).

2. The Matching Problem

Each potential pair's underlying match value combines two components: the average of the principals' individual-quality scores and complementarity between their types. Each principal's individual-quality score is the same across partners, whereas complementarity depends on the pairing. The parameter ρ controls the relative contributions of these components. The parameter δ makes complementarity harder to predict by increasing the difference between otherwise similar pairs on opposite sides of a boundary in type-pair space. A pair's underlying match value does not depend on which principal initiates the match. Both principals observe the same realized match output.

2a. Match Output

Let q_{ij} be the realized per-period output of a match between principals i and j, measured in payoff units:

$$q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

A pair's underlying match value is $Q + f(\mathbf{x}_i, \mathbf{x}_j)$, determined by the two types and the matching-function objects fixed at initialization. Because the shock ε_{ij} is independent and has mean zero, underlying match value also equals $E[q_{ij} \mid \mathbf{x}_i, \mathbf{x}_j]$, holding the matching environment fixed. The shock creates variation in realized match output that neither the principals nor the broker can predict.

The function $f : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is unknown to all principals:

$$f(\mathbf{x}_i, \mathbf{x}_j) = \frac{1}{s_{\text{mix}}(\rho, \delta)} \left[\sqrt{\rho}\, \widetilde{U}_{ij} + \sqrt{1-\rho}\, \widetilde{W}_{ij}(\delta) \right].$$

The tildes indicate that the individual-quality component U_{ij} (§2b) and the complementarity component $W_{ij}(\delta)$ (§2c) are centered and scaled using fixed moments. These fixed moments are calculated across all unordered pairs of distinct principals within each of 50 independently generated calibration markets of 1,000 principals, then averaged with equal weight across markets. The resulting coefficients depend on ρ and δ, are shared across simulations, and remain fixed through turnover.

Let $\kappa(\delta)$ denote the average within-market covariance between the standardized individual-quality and complementarity components in the calibration markets. The common adjustment

$$s^2_{\text{mix}}(\rho, \delta) = 1 + 2\sqrt{\rho(1-\rho)}\, \kappa(\delta)$$

accounts for this covariance. Each standardized component and the combined systematic component f have unit average within-market variance in the calibration markets for every ρ and δ. Realized within-market variance of f can differ from one and change with turnover. The calibration normalization defines the payoff unit. In these units, $Q = 2.84$ and $\sigma_\varepsilon = 0.28$.

The standardized components have equal average within-market variances across the calibration markets. On this calibration scale, after weighting, the individual-quality component accounts for a share ρ of their summed marginal variances, and complementarity accounts for $1 - \rho$. At $\rho = 0$, match value depends only on complementarity; at $\rho = 1$, it depends only on individual quality. At $\rho = 1$, δ has no effect.

2b. Individual Quality

The vector $\mathbf{c}$ defines a common ideal type and is fixed within each simulation. Principal i's individual-quality score, $\mathbf{x}_i^\top \mathbf{c}$, measures its alignment with that ideal. A pair's individual-quality component is the average of the two principals' scores:

$$U_{ij} = \frac{1}{2}(\mathbf{x}_i^\top \mathbf{c} + \mathbf{x}_j^\top \mathbf{c}), \qquad \widetilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}.$$

The ideal type $\mathbf{c} \in \mathbb{R}^d$ is drawn at initialization as a perturbation of a random point on the principal type curve:

$$u_c \sim \text{Unif}[0,1], \qquad \boldsymbol{\epsilon}_c \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d}\mathbf{I}_d\right), \qquad \mathbf{c} = \frac{\boldsymbol{\gamma}(u_c) + \boldsymbol{\epsilon}_c}{\|\boldsymbol{\gamma}(u_c) + \boldsymbol{\epsilon}_c\|_2}.$$

The normalization places $\mathbf{c}$ on the same unit sphere as principal types. The perturbation therefore changes the ideal direction without changing its magnitude.

2c. Match-Specific Complementarity

Define base complementarity, signed complementarity, and difficulty-adjusted complementarity as

$$V_{ij} = \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j, \qquad H_{ij} = \text{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j) V_{ij}, \qquad W_{ij}(\delta) = V_{ij} + \delta H_{ij}.$$

The normalized complementarity used in the matching function is

$$\widetilde{W}_{ij}(\delta) = \frac{W_{ij}(\delta) - \mu_W(\delta)}{\sigma_W(\delta)}.$$

A is a symmetric positive definite matrix that gives a smooth bilinear complementarity geometry. Because V_{ij} depends jointly on both types, it cannot in general be represented as the sum of separate principal scores. A second symmetric matrix, **B**, assigns pairs to two regions of type-pair space. Both matrices are fixed for the duration of the simulation.

Base complementarity. Draw $\mathbf{M}_A \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0,1)$ entries and set

$$\mathbf{A} = \mathbf{M}_A^\top \mathbf{M}_A \cdot \frac{d}{\text{tr}(\mathbf{M}_A^\top \mathbf{M}_A)}.$$

Thus **A** is symmetric positive definite and $\text{tr}(\mathbf{A}) = d$.

The trace normalization prevents arbitrary scale differences in **A** across realizations. The fixed pair-level normalization described in §2a then places complementarity on the common scale used in the matching function.

Difficulty parameter. Equivalently, write $W_{ij}(\delta) = g(\mathbf{x}_i, \mathbf{x}_j) V_{ij}$, where a second symmetric matrix $\mathbf{B} \in \mathbb{R}^{d \times d}$ defines

$$g(\mathbf{x}_i, \mathbf{x}_j) = 1 + \delta \cdot \text{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j)$$

where $\delta \in [0, 1]$ (default 0.5).

The multiplier is $1 + \delta$ when $\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j > 0$, $1 - \delta$ when it is negative, and 1 when it is zero. At $\delta = 0$, the boundary between the two regions has no effect. For $\delta > 0$, pairs on opposite sides of the boundary can have sharply different underlying match values even when their types are otherwise similar. Increasing δ increases this difference, making complementarity harder to learn. The scale $\sigma_W(\delta)$ holds the average within-market variance of standardized complementarity at one in the calibration markets as δ varies.

Boundary construction. At initialization, **B** is made orthogonal to **A** under the weighted inner product defined by the initial principals' empirical second-moment matrix. The resulting boundary remains fixed through turnover. Let

$$\mathbf{\Sigma}_x = \frac{1}{N} \sum_{i=1}^{N} \mathbf{x}_i \mathbf{x}_i^\top$$

be the uncentered second-moment matrix of the initial type vectors.

Draw $\mathbf{Z} \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0,1)$ entries and form a symmetric zero-trace matrix:

$$\widetilde{\mathbf{B}} = \frac{\mathbf{Z} + \mathbf{Z}^\top}{2}, \qquad \mathbf{B}_0 = \widetilde{\mathbf{B}} - \frac{\operatorname{tr}(\widetilde{\mathbf{B}})}{d}\mathbf{I}_d.$$

Define the weighted inner product

$$\langle \mathbf{M}, \mathbf{N} \rangle_{\boldsymbol{\Sigma}_x} = \operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{M} \boldsymbol{\Sigma}_x \mathbf{N}),$$

then remove the projection of $\mathbf{B}_0$ onto $\mathbf{A}$ and normalize:

$$\mathbf{B}_{\text{raw}} = \mathbf{B}_0 - \frac{\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{B}_0 \boldsymbol{\Sigma}_x \mathbf{A})}{\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{A})}\mathbf{A}, \qquad \mathbf{B} = \frac{\mathbf{B}_{\text{raw}}}{\|\mathbf{B}_{\text{raw}}\|_F}.$$

This gives $\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{B} \boldsymbol{\Sigma}_x \mathbf{A}) = 0$ and $\|\mathbf{B}\|_F = 1$. Weighting by $\boldsymbol{\Sigma}_x$ removes overlap between the two bilinear scores under the initial empirical type distribution, rather than in unused regions of $\mathbb{R}^d$.

Symmetry of $\mathbf{A}$ and $\mathbf{B}$ makes $f(\mathbf{x}_i, \mathbf{x}_j) = f(\mathbf{x}_j, \mathbf{x}_i)$.

3. Learning

Before searching for matches, principals and the broker may update their prediction models on their accumulated histories. Updated models are used to score candidates (§6). The broker is eligible to refit every period. Principal i is eligible when $i \bmod 2 = t \bmod 2$, so each principal is eligible every other period. An eligible model refits only when its history is nonempty and it has received new observations since its previous fitting call. A parameter switch selects the prediction model. Neural network (NN) learning is the default. Ridge regression is the alternate experiment.

3a. Prediction Models and Fitting

In the default model, both principals and the broker use a fully connected network with one hidden layer, ReLU activations, and a single linear output:

$$\hat{q}(\mathbf{z}) = \mathbf{w}_2^\top \operatorname{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + b_2$$

where $\hat{q}$ is predicted match value, $\mathbf{z}$ is the input feature vector and the remaining symbols are network weights and biases. Principals receive raw partner type vectors; the broker receives pairs of them.

Training rule. Network weights minimize mean squared error (MSE) with the Adam optimizer on the current training window. Principals and the broker may have separate learning rates, $\eta_{\text{lr},a}$ and $\eta_{\text{lr},b}$. No explicit weight regularization is applied. At $t = 0$, each network sets $\mathbf{W}_1$ using He initialization, $\mathbf{b}_1 = \mathbf{0}$, $\mathbf{w}_2 = \mathbf{0}$, and $b_2 = Q$ (§2a). An untrained predictor therefore returns exactly Q for every input. Networks with seed observations then take $E_{\text{init},c}$ gradient steps on their seed histories, where $c \in \{a, b\}$.

Alternate ridge rule. In the ridge experiment, principals and the broker instead use

$$\hat{q}_i(\mathbf{x}_j) = \alpha_i + \boldsymbol{\beta}_i^\top \mathbf{x}_j, \qquad \hat{q}_b(\mathbf{x}_i, \mathbf{x}_j) = \alpha_b + \boldsymbol{\gamma}_b^\top \psi(\mathbf{x}_i, \mathbf{x}_j).$$

Principal and broker fits minimize mean squared error plus λ_a and λ_b, respectively, times the squared Euclidean norm of the slope coefficients. Intercepts are not penalized. Raw type and pair

features are used without sample standardization. Before its first fit, a ridge predictor has zero slopes and intercept Q. The reporting simulations use $\lambda_a = 0.001$ and $\lambda_b = 0.001$.

Subsequent NN updates warm-start from the previous weights and persistent Adam moment state. The number of gradient steps adapts to the ratio of new observations to total history:

$$E_{c,t} = \max\left(E_{\text{steps},c},\ \left\lceil E_{\text{init},c} \cdot \frac{n_{\text{new}}}{n_{\text{total}}} \right\rceil\right), \qquad c \in \{a, b\},$$

where n_{new} is the number of observations added since that model's previous fitting call, $n_{\text{total}} = |\mathcal{H}^t|$ is full history size, and $E_{\text{steps},c}$ is the learner-specific minimum per-call step count. For a principal, n_{new} can span multiple periods because principals fit on an alternating schedule. Principals and the broker use the same functional rule but may have different parameter values. Ridge is solved directly at each fitting call. The reporting NN values are $\eta_{\text{lr},a} = 0.003$, $\eta_{\text{lr},b} = 0.03$, $E_{\text{init},a} = E_{\text{init},b} = 100$, and $E_{\text{steps},a} = E_{\text{steps},b} = 50$.

Initialization is period 0 for each learner. Its seed-history count is stored as the first period boundary. Each later boundary stores cumulative history at the end of a simulation period. A fitting call uses observations from the most recent $W_p = 40$ completed simulation periods. Before 40 simulation periods are available, the window also includes period 0. Once 40 are available, the period-0 seed history is excluded. If the selected history block exceeds $M_{\text{train}} = 2000$ observations, fitting deterministically keeps 2000 spaced observations across that block, including its oldest and newest observations. If the cap is one, it keeps the newest observation. The cap changes only the fitting sample, not the stored history or the period window. NN and ridge use the same window, cap, fitting cadence, and histories.

3b. Principal i's Model

History. $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}_{m=1}^{n_i}$ records the other party's type and realized match output from every match involving i.

Input. The principal's network takes the partner's type as input: $\mathbf{z} = \mathbf{x}_j$.

The hidden width is $h_A = 2d$, so the baseline uses 16 hidden units.

3c. Broker's Model

History. $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}_{m=1}^{n_b}$ records both parties' types and realized match output from broker-mediated matches.

Input. The broker's network takes a symmetric pair feature vector:

$$\psi(\mathbf{x}_i, \mathbf{x}_j) = \left[\mathbf{x}_i + \mathbf{x}_j;\ \text{vech}\left(\frac{\mathbf{x}_i\mathbf{x}_j^\top + \mathbf{x}_j\mathbf{x}_i^\top}{2}\right)\right].$$

The half-vectorization vech keeps the lower-triangular entries of the symmetric matrix, so the broker input dimension is $d_B = d + d(d+1)/2$, equal to 44 at $d = 8$.

The broker hidden width is $h_B = 8d$, so the baseline uses 64 hidden units.

Because $\psi(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_j, \mathbf{x}_i)$, each observed unordered pair contributes one training row.

The broker learns only from mediated matches. It does not observe self-search outcomes.

3d. Ridge Broker Ablations

Three separate ridge-only ablations change the broker while leaving every principal's local ridge model unchanged. **Size-matched pooled history** uniformly samples the broker's windowed and capped fitting data without replacement, using the smaller of its available sample size and the nearest-integer median sample size among principals with nonempty histories, with half-integers rounded up. **Single-principal history** randomly retains one endpoint when each broker observation is recorded, fits $g_b(\mathbf{x}) = \alpha_b + \boldsymbol{\gamma}_b^\top \mathbf{x}$, and scores a pair as $g_b(\mathbf{x}_i) + g_b(\mathbf{x}_j) - \bar{q}_b$, where $\bar{q}_b$ is the current fitting-sample output mean and defaults to Q before fitting. **Additive ridge** uses $\mathbf{x}_i + \mathbf{x}_j$ without bilinear features. All use the calibrated principal and broker penalties, period window, observation cap, and fitting cadence.

4. Match Economics

Matches use a common reservation threshold and channel-specific cost accounting.

At initialization, each simulation computes $\bar{q}_{\text{cal}}$ as the sample mean of $Q + f(\mathbf{x}_i, \mathbf{x}_j)$ over 10,000 pairs whose indices are drawn independently and uniformly with replacement from the initial principals. This quantity scales the reservation threshold, broker fee, and self-search cost.

4a. Outside Options

All principals share a reservation threshold r, calibrated at initialization:

$$r = f_r \cdot \bar{q}_{\text{cal}}$$

where $\bar{q}_{\text{cal}}$ is the calibration mean and f_r is the reservation coefficient (default 0.60). Since ϕ and c_s do not depend on r (§4c), f_r may take any nonnegative value, including $f_r \geq 1$, and is swept as an analysis axis.

4b. Participation Constraints

Demanders emit offers only when the candidate evaluation exceeds r. In self-search, this evaluation is the pair's mean past realized output $\bar{q}_{ij}$ for neighbors with prior pair-specific experience and the demander prediction $\hat{q}_i(\mathbf{x}_j)$ otherwise (§6a). Brokered search uses the broker pair prediction $\hat{q}_b(\{i, j\})$. A one-sided offer is accepted only if the receiver's evaluation exceeds r; reciprocal offers are accepted automatically (§6d).

4c. Search Costs

The self-search cost and broker fee share one calibrated level:

$$\phi = c_s = \lambda_c \cdot \bar{q}_{\text{cal}}.$$

λ_c is the shared cost rate (default 0.05). The two symbols differ in payoff accounting:

- **Self-search** costs c_s on each requested self-search position, whether or not it is filled.
- **Brokerage** charges ϕ only on successful brokered placements.

5. Network Structure and Turnover

Principals interact through a single undirected network G that determines search opportunities and structural position.

5a. Network Initialization

G is initialized as a Watts-Strogatz small-world graph on the N principals (Watts & Strogatz, 1998), with ring degree $k_G = 6$, rewiring probability $p_{\text{rewire}} = 0.1$, and random principal order on the ring.

The broker is added as a permanent node with no type vector. Broker-node edges are synchronized to standing roster members, current-period broker clients, and principals currently engaged in broker-channel matches (§6b).

5b. Match Tie Formation

Each realized match adds an undirected i-j edge if one does not already exist. Ties persist until either node exits.

5c. Principal Turnover

Principals exit independently each period with probability η (default 0.02). An exiting principal is removed everywhere. Other actors retain their accumulated observations that involve it. Its node slot is reused by an entrant with a fresh type drawn from the same type process (§1). The entrant receives $\lfloor k_G/2 \rfloor$ edges to existing principals sampled with probability proportional to $\exp(-\|\mathbf{x}_{i'} - \mathbf{x}_j\|^2)$.

6. Search

At the start of each period, each principal draws active relationship demand

$$d_i \sim \text{Binomial}(K,\ p_{\text{demand}}),$$

where K is the maximum number of active demands the principal can initiate in the period and p_{demand} defaults to 0.50. If $d_i > 0$, the principal chooses **one channel for the batch** (§7): self-search or broker.

6a. Self-Search

Principal i's self-search candidate pool has two components:

Neighbors. For a direct network neighbor j with prior pair-specific experience, principal i uses $\bar{q}_{ij} = \frac{1}{n_{ij}} \sum q_{ij}^{(m)}$, where n_{ij} is the number of times i and j have matched. If the graph edge exists but $n_{ij} = 0$, the principal instead uses $\hat{q}_i(\mathbf{x}_j)$. Initial histories seed one observation for every initial graph edge. Unobserved neighbor edges arise after turnover because entrants attach structurally before matching.

Strangers. Each self-searching demander i independently samples a pool U_i^t of $\min(n_{\text{strangers}}, N)$ principals uniformly without replacement, where $n_{\text{strangers}} = 10$ (default). The demander evaluates pool members that are not current neighbors using $\hat{q}_i(\mathbf{x}_j)$ (§3b).

Principal i orders feasible self-search candidates by these evaluations and emits up to d_i directed offers to candidates above r. Equal evaluations are ordered uniformly at random, so a principal whose predictions are constant produces a random ranking rather than an undefined ordering.

6b. Broker Roster and Access

The broker maintains a **roster** of principals it can propose as counterparties when mediating matches. In each period, it also observes the current-client set D^t of principals who outsourced. Broker access for the period is:

$$\mathcal{A}_b^t = \text{Roster}^t \cup D^t.$$

Current clients expand broker access for the period but do not become standing roster members.

Initialization. The roster is seeded with a target size

$$R^* = \lceil \alpha_R N \rceil,$$

where the broker roster share α_R defaults to 0.20. The model draws R^* principals uniformly at random (default 200 at $N = 1000$). The broker's history is seeded from existing roster-roster ties in the initial graph, capped at 100 observed ties.

Roster refresh. At the start of each period, each member independently leaves the roster with probability p_{roster} (default 0.02). If $\widetilde{\text{Roster}}^t$ is the post-churn roster,

$$\widetilde{\text{Roster}}^t = \{i \in \text{Roster}^{t-1} : u_i^t > p_{\text{roster}}\}, \qquad u_i^t \overset{iid}{\sim} \text{Unif}[0,1],$$

then the broker samples without replacement from $\{1, \ldots, N\} \setminus \widetilde{\text{Roster}}^t$ until $|\text{Roster}^t| = R^*$, or until the population is exhausted. Broker-node edges are synchronized to the standing roster, current clients, and current broker-mediated matches (§5a).

6c. Broker-Mediated Search

For every unordered pair $\{i, j\}$ such that one side is a broker-client demander and the other side is in $\mathcal{A}_b^t$, the broker computes predicted match value. For each broker-client demander i, it ranks accessible counterparties above r, breaking equal predictions uniformly at random, and keeps the highest-ranked candidates up to i's residual active broker demand. Directed broker offers are then processed in a global score ranking with a new, independent random ordering within score ties. If both sides are broker-client demanders with remaining active demand, the same unordered pair can generate reciprocal broker offers.

Assessment-only. The broker scores the principal's network neighbors and an independently sampled stranger pool (§6a) using its pair predictions.

Access-only. Candidates come from the broker access set $\mathcal{A}_b^t$. Scores use the requesting principal's pair-specific mean for experienced neighbors and its prediction model otherwise (§6a).

All other rules remain as specified. The broker still records mediated matches and updates its prediction model (§3), but its predictions do not determine access-only offers.

6d. Offer Acceptance and Match Formation

All self-search and broker offers enter one shared market, grouped by unordered pair.

1. If pair $\{i, j\}$ contains reciprocal offers $i \to j$ and $j \to i$, the relationship is accepted automatically.
2. If pair $\{i, j\}$ contains one directed offer $i \to j$, receiver j evaluates i using $\bar{q}_{ji}$ for known partners and $\hat{q}_j(\mathbf{x}_i)$ otherwise. The offer is accepted iff this value exceeds r.
3. Each accepted unordered pair forms at most one current-period relationship.

At market finalization, for each accepted match: 1. Realized output is drawn: $q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}$. 2. Both parties add the observation to their histories: i adds $(\mathbf{x}_j, q_{ij})$ to $\mathcal{H}_i$; j adds $(\mathbf{x}_i, q_{ij})$ to $\mathcal{H}_j$. 3. If either directed offer used the broker channel, the broker adds $(\mathbf{x}_i, \mathbf{x}_j, q_{ij})$ to $\mathcal{H}_b$ once. 4. An edge is added between i and j in G (if not already present).

A demander can have at most d_i outgoing offers, but can receive any number of acceptable incoming offers. Thus K is an active-demand bound, not a capacity constraint on receiving offers.

Before the next period begins, match lists are cleared. The next period begins with no current-period matches.

7. The Outsourcing Decision

7a. Satisfaction Tracking

Each principal i maintains a satisfaction index $s^t_{i,c}$ for each search channel $c \in \{\text{self}, \text{broker}\}$. Satisfaction and reputation are indexed by the period just completed, with initialization indexed by 0. For the chosen channel, the index updates once per active-demand period as an exponentially weighted moving average (EWMA) of realized match output net of search costs:

$$s^t_{i,c} = (1 - \omega)\, s^{t-1}_{i,c} + \omega \cdot \tilde{q}$$

where $\omega = 0.2$ and $\tilde{q}$ is the period-t input in the table. The denominator is requested demand d_i, so unfilled requests contribute zero output. A reciprocal relationship can update two channel-specific states, one for each directed offer.

Channel	Satisfaction input $\tilde{q}$
Self-search	$\dfrac{\sum q_{ij}}{d_i} - c_s$, summing over accepted directed offers made by i through self-search
Brokered	$\dfrac{\sum q_{ij} - \phi \cdot n_{i,\text{broker success}}}{d_i}$, summing over accepted directed offers made by i through the broker

Brokered failure gives $\tilde{q} = 0$; self-search failure gives $\tilde{q} = -c_s$.

$s^t_{i,\text{self}}$ covers all self-search, including known neighbors.

At initialization, satisfaction with self-search is the mean of seed match outcomes and satisfaction with brokerage is the broker's seed-data reputation (§7c). Fresh entrants set satisfaction with self-search to the neighbors' mean for that channel and satisfaction with brokerage to current broker reputation.

The first broker choice, regardless of placement success, switches the principal from broker reputation to its own $s^t_{i,\text{broker}}$ (§7b).

7b. Decision Rule

At the start of period t, each principal with $d_i > 0$ compares:

- $\textbf{score}_{\textbf{self}} = s^{t-1}_{i,\text{self}}$
- $\textbf{score}_{\textbf{broker}} = s^{t-1}_{i,\text{broker}}$ if the principal has tried the broker, otherwise rep^{t-1}_b.

The principal chooses the higher-score channel. Ties are resolved by a fair coin flip.

7c. Broker Reputation

$$\text{rep}^t_b = \begin{cases} \frac{1}{|D^t|} \sum_{i \in D^t} s^t_{i,\text{broker}} & \text{if } D^t \neq \emptyset \\ \text{rep}^{t-1}_b & \text{otherwise} \end{cases}$$

where D^t is the set of principals who outsourced to the broker this period. With current clients, reputation is their mean post-update broker satisfaction; with no clients, it is unchanged. Reputation is initialized from the mean broker seed outcome.

8. Period Measures

Period measures are evaluated after current-period matching, satisfaction, and reputation updates, but before principal entry/exit turnover. Principal decisions do not depend on these measures.

Network Measures

Broker network-position measures use G, including the broker node (§5a). They are computed when $t \bmod \Delta_{\text{net}} = 0$ (default 20). Other period measures are recorded each period.

Betweenness centrality. Freeman betweenness (Freeman, 1977) for the broker node:

$$\text{BC}_b = \frac{1}{(n-1)(n-2)} \sum_{u \neq b} \sum_{v \notin \{u,b\}} \frac{\sigma_{uv}(b)}{\sigma_{uv}}$$

where $n = N + 1$, σ_{uv} is the number of shortest paths from u to v, and $\sigma_{uv}(b)$ is the number passing through b. Pairs with no connecting path contribute zero. The double sum counts undirected pairs from both directions, so the denominator is $(n-1)(n-2)$ (Brandes, 2001, p. 172).

Burt's constraint. Broker ego-network constraint (Burt, 1992):

$$\text{Constraint}_b = \sum_{j \in \mathcal{N}_G(b)} \left(p_{bj} + \sum_{h \in \mathcal{N}_G(b) \setminus \{j\}} p_{bh}\, p_{hj} \right)^2$$

where $\mathcal{N}_G(u)$ is node u's set of neighbors in G, $d_u = |\mathcal{N}_G(u)|$, and $p_{uv} = 1/d_u$ when $v \in \mathcal{N}_G(u)$, and zero otherwise (Everett & Borgatti, 2020). The indirect term uses intermediary h's degree in G.

Effective size. Non-redundant broker contacts using the Borgatti (1997) binary-undirected simplification:

$$\text{ES}_b = d_b - \frac{2t_b}{d_b},$$

where d_b is the broker's degree and t_b is the number of ties among the broker's neighbors, excluding ties to the broker.

Prediction Quality

Prediction quality is evaluated on random candidate pairs:

Sample	Construction	Reported measures
Random candidate pairs	Up to 100 principals with nonempty histories; for each sampled principal i, $\min(40, N-1)$ non-self partners j. Each predicted match value is evaluated against $Q + f(\mathbf{x}_i, \mathbf{x}_j)$ for the same sampled pair. Both models use the same sample, which is deterministic conditional on seed and period.	Per-principal R^2, root mean squared error (RMSE), bias, and Spearman rank correlation, averaged across sampled principals.

The definitions are

$$R^2 = 1 - \frac{\text{MSE}}{\text{Var}_{\text{pop}}(q)}, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum(\hat{q} - q)^2}, \qquad \text{Bias} = \frac{1}{n}\sum(\hat{q} - q).$$

Rank correlation evaluates the ranking produced from predicted match value. Candidates are ordered by predicted match value, with equal predictions ordered uniformly at random. All tie-breaking draws are independent. For random candidate pairs, Spearman's ρ_S compares this produced ranking with ranks of underlying match value. Thus a constant predictor would produce a random ranking with expected rank correlation zero. In the definitions above, q denotes the relevant target: underlying match value for random candidate pairs. Random-candidate summaries exclude a principal's candidate set from all four measures if it contains fewer than five pairs or if the population variance of its underlying match values is below $\sigma_\varepsilon^2/6$. The R^2, RMSE, bias, and rank-correlation means therefore use the same valid-principal sample.

Other Measures

Measure	Definition
Access vs. assessment	For each accepted broker-directed offer, record whether receiver j was a direct neighbor of sender i in G at the time of the offer. Non-neighbor receivers indicate access; neighbor receivers indicate assessment.
Realized match output by channel	Average realized match output $\bar{q}_c^t$, where $c \in \{\text{self}, \text{brokered}\}$.
Requested-position fill rates	For each channel, accepted directed offers divided by requested positions.
Net output per principal	Total realized output from completed relationships, counting each relationship once, minus all self-search costs and broker fees, divided by N. Self-search costs apply to every self-search request, successful or not. Broker fees apply to each accepted broker-directed offer, including both offers when a reciprocal relationship fulfills two brokered requests.
Mean channel satisfaction	Cross-principal means $N^{-1}\sum_i s_{i,\text{self}}^t$ and $N^{-1}\sum_i s_{i,\text{broker}}^t$.
Outsourcing rate	Outsourced requested positions divided by total requested positions.
Broker access size	$\lvert\mathcal{A}_b^t\rvert = \lvert\text{Roster}^t \cup D^t\rvert$, measured after outsourcing decisions form D^t and before entry/exit.
Counterparty concentration	Median and maximum current-period counterparties per principal.
Principal-node degree summaries	Mean, median, minimum, and maximum degree across principals.

For channel accounting, each completed relationship's output is counted once per period and assigned to brokerage if at least one accepted offer used the broker; otherwise it is assigned to self-search. Brokered net output subtracts broker fees from brokered realized output. Self-search net output subtracts all self-search costs from self-search realized output. Dividing each channel's net output by N gives its net output per principal.

Part II. Baseline Values and Reproducibility

9. Baseline Values and Reproducibility

9a. Baseline Parameter Values

The table lists baseline values only. Entries are arranged in two columns; each column follows the document order, with the run length listed first.

The baseline learning function is NN. Ridge and its broker variants are alternate experiments (§3).

The baseline broker provides both access and assessment. The assessment-only and access-only modes are alternate experiments (§6c).

Symbol	Baseline	Defined in	Symbol	Baseline	Defined in
T	500	§1	f_r	0.60	§4a
N	1000	§1	λ_c	0.05	§4c
d	8	§1	k_G	6	§5a
d_γ	8	§1	p_{rewire}	0.1	§5a
σ_x	0.5	§1	η	0.02	§5c
Q	2.84	§2a	K	5	§6
ρ	0.50	§2a	p_{demand}	0.50	§6
δ	0.5	§2c	$n_{\text{strangers}}$	10	§6a
σ_ε	0.28	§2a	α_R	0.20	§6b
$\eta_{\text{lr},a}, \eta_{\text{lr},b}$	0.003, 0.03	§3a	p_{roster}	0.02	§6b
$E_{\text{init},a}, E_{\text{init},b}$	100, 100	§3a	ω	0.2	§7a
$E_{\text{steps},a}, E_{\text{steps},b}$	50, 50	§3a	Δ_{net}	20	§8
W_p	40 periods	§3a	M_{train}	2000	§3a

9b. Reproducibility

All model-event randomness flows from a single integer seed. The seed determines type draws, G, matching function parameters ($\mathbf{c}$, $\mathbf{A}$, $\mathbf{B}$), broker seed roster, standing-roster churn and replenishment, random ordering within market score ties, and subsequent model events. Random candidate samples and their independent principal and broker tie-breaking draws are deterministic conditional on seed and period. Simulations are fully reproducible given parameters and seed.

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